

DEG Difference Analysis Report

I have analyzed Vera Yu's differential expression gene (DEG) results reported in the `Table S1. DEGs` tab of `ALZ-22-e71463-s002.xlsx`, which is provided as one of the supplemental information of the paper. This tab contains 118,297 DEG data rows. Each row represents a gene–cell-type–contrast DEG result.

Row number breakdown by contrast

Contrast	Interpretation	Rows	Percent
F_e2x_AD_vs_F_e2x_NCI	Female, APOE ε2 carrier, AD vs NCI	10,952	9.26%
F_e33_AD_vs_F_e33_NCI	Female, APOE ε3/ε3, AD vs NCI	11,236	9.50%
F_e4x_AD_vs_F_e4x_NCI	Female, APOE ε4 carrier, AD vs NCI	20,942	17.70%
M_e2x_AD_vs_M_e2x_NCI	Male, APOE ε2 carrier, AD vs NCI	45,781	38.70%
M_e33_AD_vs_M_e33_NCI	Male, APOE ε3/ε3, AD vs NCI	13,120	11.09%
M_e4x_AD_vs_M_e4x_NCI	Male, APOE ε4 carrier, AD vs NCI	16,266	13.75%
Total		118,297	100%

Yu's spreadsheet contains only DEG rows satisfying both of the following criteria:

- absolute fold change greater than 1.3 ($\text{abs}(\text{avg_log2FC}) > \log_2(1.3)$); and
- adjusted p-value less than 0.05 ($\text{p_val_adj} < 0.05$).

Comparison with my DEG results

I can only compare my results with the DEG rows provided in Yu's spreadsheet. Because the spreadsheet does not include Yu's non-DEG results, a row absent from the spreadsheet cannot be compared directly at the raw-statistic level.

In this comparison, “my side” includes only my DEG rows that meet the same thresholds: absolute fold change greater than 1.3 ($\text{abs}(\log\text{FC}) > \log_2(1.3)$) and adjusted p-value less than 0.05 ($\text{fdr_bh_within_contrast} < 0.05$). Rows were matched using the combination of cell type, contrast, and gene.

Contrast	Yu DEG rows	My DEG rows	Overlapping rows	In Yu but missing from my side	Additional from my side
F_e2x_AD_vs_F_e2x_NCI	10,952	10,698	10,002	950	696
F_e33_AD_vs_F_e33_NCI	11,236	11,130	10,999	237	131
F_e4x_AD_vs_F_e4x_NCI	20,942	20,029	18,804	2,138	1,225
M_e2x_AD_vs_M_e2x_NCI	45,781	41,681	39,919	5,862	1,762
M_e33_AD_vs_M_e33_NCI	13,120	13,809	12,924	196	885
M_e4x_AD_vs_M_e4x_NCI	16,266	14,254	13,951	2,315	303
Total	118,297	111,601	106,599	11,698	5,002

“In Yu but missing from my side” means a DEG row reported by Yu that is not called as a DEG in my results.
“Additional from my side” means a DEG row called in my results that is absent from Yu’s DEG spreadsheet.

Adjusted p-value comparison for all Yu DEG rows

All 118,297 rows in Yu’s DEG spreadsheet have exactly one corresponding tested row in my Phase 08 results, matched by cell type, contrast, and gene. This comparison uses the adjusted p-value from every matched row on my side, including rows where my adjusted p-value does not pass the 0.05 DEG threshold.

Yu DEG rows	Matched rows on my side	Missing adjusted p-values on my side	My adjusted p < 0.05	My adjusted p >= 0.05
118,297	118,297	0	106,599	11,698

Absolute adjusted p-value difference

I calculated the absolute adjusted p-value difference for every matched row:

`absolute difference = abs(my adjusted p-value - Yu adjusted p-value)`

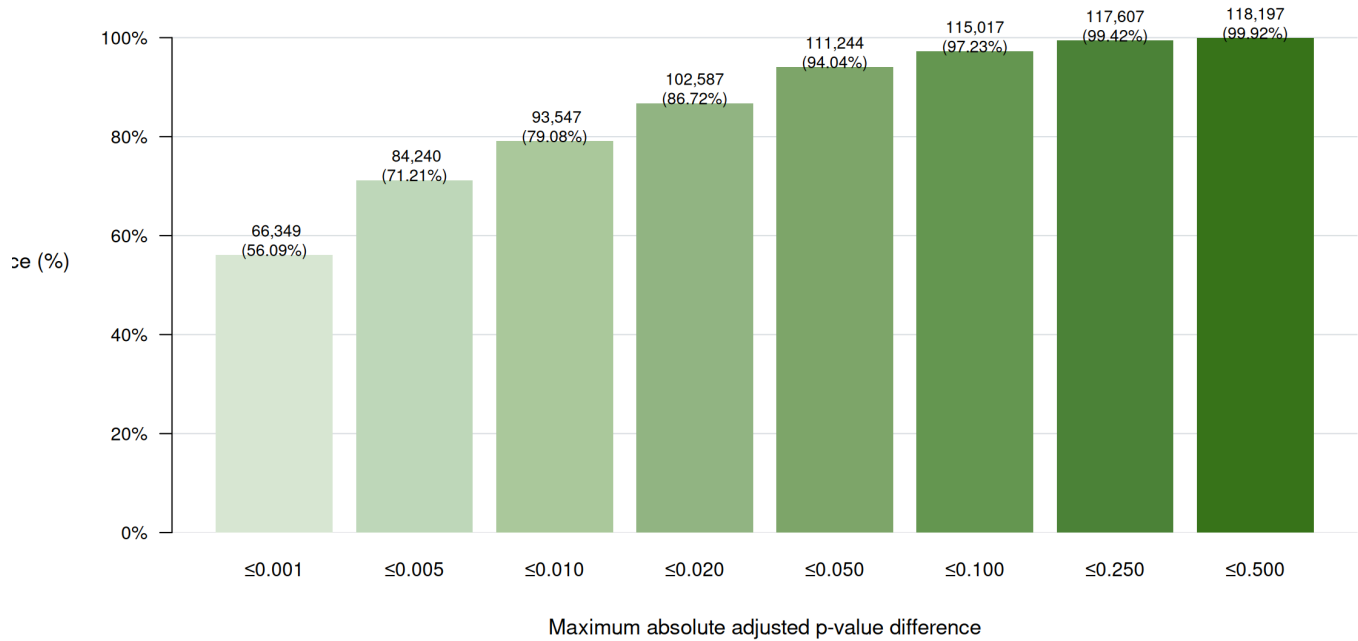
The absolute difference remains defined when either adjusted p-value is zero. The counts below are cumulative.

Maximum absolute difference	Rows within tolerance	Percent of all 118,297 rows
≤0.001	66,349	56.09%
≤0.005	84,240	71.21%
≤0.010	93,547	79.08%
≤0.020	102,587	86.72%
≤0.050	111,244	94.04%
≤0.100	115,017	97.23%
≤0.250	117,607	99.42%
≤0.500	118,197	99.92%

The median absolute adjusted p-value difference is 0.000422 . The 75th percentile is 0.00699 , the 90th percentile is 0.0284 , and the maximum is 0.980 . Overall, 79.08% of matched rows differ by no more than 0.01 in absolute adjusted p-value and 86.72% differ by no more than 0.02.

Absolute adjusted p-value agreement: my results versus Yu

Absolute difference = |my adjusted p - Yu adjusted p|



N = 118,297 matched gene–cell-type–contrast rows; tolerance counts are cumulative.

Figure 1. Number and percentage of the 118,297 matched Yu rows within each cumulative absolute adjusted p-value difference tolerance.

P-value comparison for all Yu DEG rows

All 118,297 rows in Yu's DEG spreadsheet have exactly one corresponding tested row with a finite raw p-value in my Phase 08 results. This comparison uses the raw p-value from every matched row on my side, including rows where my p-value does not pass 0.05.

Yu DEG rows	Matched rows on my side	Missing p-values on my side	My p < 0.05	My p >= 0.05
118,297	118,297	0	113,668	4,629

Absolute p-value difference

I calculated the absolute raw p-value difference for every matched row:

```
absolute difference = abs(my p-value - Yu p-value)
```

The counts below are cumulative.

Maximum absolute difference	Rows within tolerance	Percent of all 118,297 rows
≤ 0.001	83,993	71.00%
≤ 0.005	99,758	84.33%
≤ 0.010	105,611	89.28%
≤ 0.020	110,385	93.31%
≤ 0.050	114,797	97.04%
≤ 0.100	116,869	98.79%
≤ 0.250	118,054	99.79%
≤ 0.500	118,242	99.95%

The median absolute p-value difference is 0.0000524 . The 75th percentile is 0.00162 , the 90th percentile is 0.0112 , and the maximum is 0.979 . Overall, 89.28% of matched rows differ by no more than 0.01 in absolute p-value and 93.31% differ by no more than 0.02.

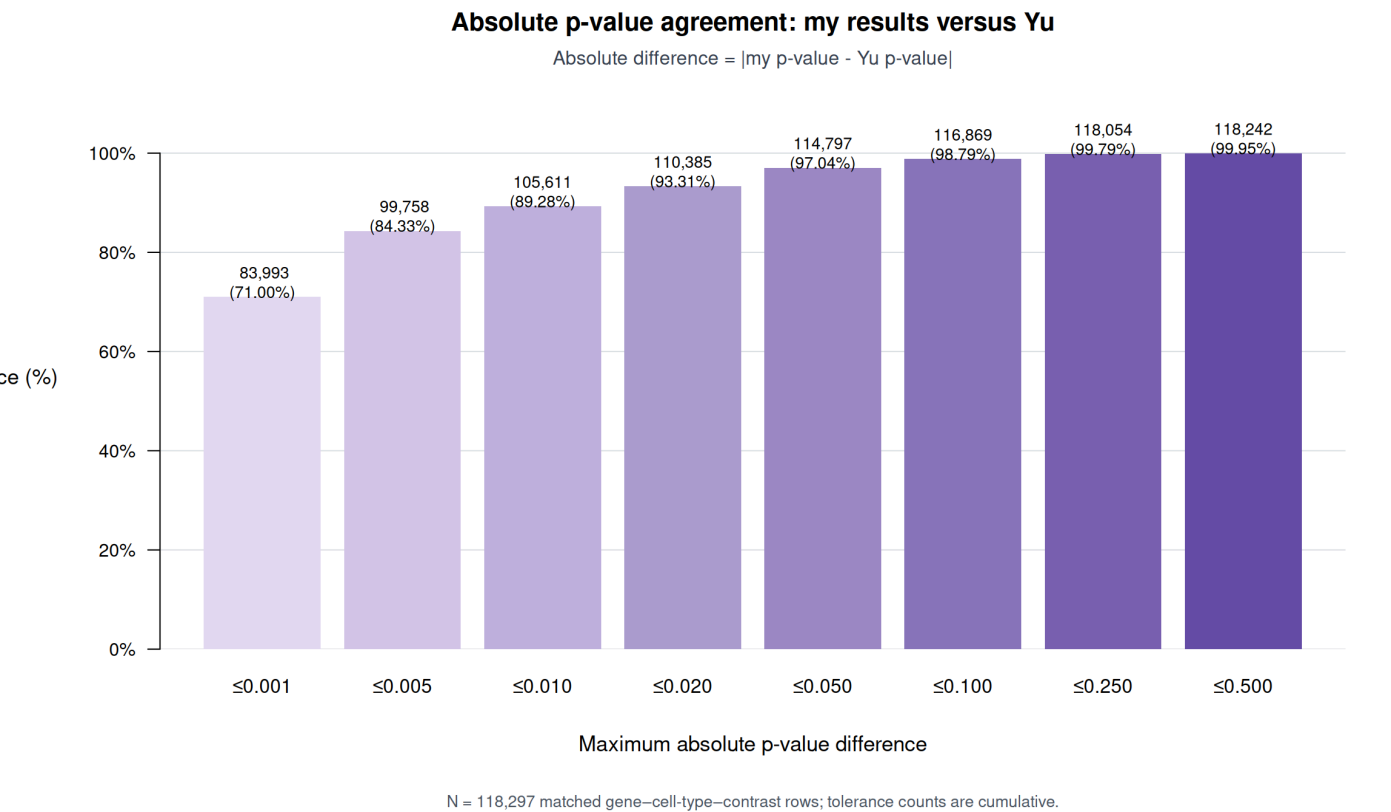


Figure 2. Number and percentage of the 118,297 matched Yu rows within each cumulative absolute raw p-value difference tolerance.

Potential source of the mismatch

1. Exact software environment

Yu reports Seurat version 5 but does not report the exact Seurat, MAST, R, Matrix, or supporting-package versions. My code used:

- R 4.3.3;
- Seurat 5.5.1;
- SeuratObject 5.4.0;
- MAST 1.28.0; and
- Matrix 1.6.5.

Differences in Seurat's MAST library version, MAST likelihood-ratio calculations, convergence handling, or supporting numerical libraries could change p-values and adjusted p-value while leaving log2FC unchanged.